# EC303 — Signals and System

## Module 2: Linear Shift-Invariant (LSI) Systems & Convolution

Prepared by **Biraj Sarkar** (CGEC) · MAKAUT B.Tech ECE (2023–27)

Authoritative Syllabus-Aligned Notes

## Contents


**1 Linear Shift-Invariant (LSI) Systems Overview . . . 1**
- 1.1 Representation of Signals Through Shifted Impulses . . . 1

**2 Impulse Response and Step Response . . . 1**
- 2.1 The Impulse Response $h(t)$ and $h[n]$ . . . 2
- 2.2 The Step Response $s(t)$ and $s[n]$ . . . 2
- 2.3 Relationship Between Impulse Response and Step Response . . . 2

**3 The Convolution Operation . . . 3**
- 3.1 Derivation of the Convolution Integral (CT) . . . 3
- 3.2 Derivation of the Convolution Sum (DT) . . . 3
- 3.3 Graphical Interpretation of Convolution . . . 4
- 3.4 Properties of Convolution . . . 5
- 3.5 Interconnection of LSI Systems . . . 5

**4 Causality and Stability of LSI Systems . . . 6**
- 4.1 Characterization of Causality . . . 6
- 4.2 Characterization of BIBO Stability . . . 6

**5 System Representation Through Differential Equations . . . 7**
- 5.1 Linear Constant-Coefficient Differential Equations (LCCDE) . . . 7
- 5.2 Complete Solution: Natural and Forced Responses . . . 7
- 5.3 Condition of Initial Rest . . . 8
- 5.4 Block Diagram Realization of LCCDE Systems . . . 8

**6 Behavior with Aperiodic Convergent Inputs . . . 8**


## 1. Linear Shift-Invariant (LSI) Systems Overview

**Definition: Linear Shift-Invariant (LSI / LTI) System**

A system $\mathcal{T}$ is called a **Linear Shift-Invariant (LSI)** or **Linear Time-Invariant (LTI)** system if it satisfies both the principle of **Linearity** (superposition: additivity and homogeneity) and the condition of **Shift-Invariance** (time-invariance).
Formally, for arbitrary scalar multipliers $a, b \in \mathbb{C}$ and arbitrary inputs $x_1(t), x_2(t)$:

1. **Superposition (Linearity):** $\mathcal{T}\{ax_1(t) + bx_2(t)\} = a\mathcal{T}\{x_1(t)\} + b\mathcal{T}\{x_2(t)\} =$

$ay_1(t) + by_2(t)$.

2. **Shift-Invariance (Time-Invariance):** If $\mathcal{T}\{x(t)\} = y(t)$, then $\mathcal{T}\{x(t-t_0)\} = y(t-t_0)$ for any arbitrary shift $t_0 \in \mathbb{R}$ (or $n_0 \in \mathbb{Z}$ in discrete-time).

### 1.1 Representation of Signals Through Shifted Impulses

Any continuous-time signal $x(t)$ can be viewed as a continuum of weighted, shifted Dirac delta impulses via the sifting property:

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \tag{1}$$

Similarly, any discrete-time signal $x[n]$ is exactly decomposed into a linear combination of weighted, shifted Kronecker deltas:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \tag{2}$$

This decomposition forms the conceptual bedrock of LTI system analysis: because the system is linear and time-invariant, knowledge of its response to a single unit impulse ($\delta(t)$ or $\delta[n]$) is sufficient to completely determine its response to *any arbitrary* input signal.

## 2. Impulse Response and Step Response

### 2.1 The Impulse Response $h(t)$ and $h[n]$

**Definition: Impulse Response**

The **Impulse Response** $h(t)$ (in CT) or $h[n]$ (in DT) is the output of an LSI system when the input is a unit impulse applied at the origin under initial rest conditions (zero energy stored in the system):

$$h(t) \triangleq \mathcal{T}\{\delta(t)\} \quad \text{(Continuous-Time)} \tag{3}$$
$$h[n] \triangleq \mathcal{T}\{\delta[n]\} \quad \text{(Discrete-Time)} \tag{4}$$

**Physical Significance:**

- It acts as the "fingerprint" or signature of an LSI system. Every structural attribute—memory, causality, stability, invertibility, and frequency selectivity—is entirely encoded in $h(t)$ or $h[n]$.
- It characterizes the transient relaxation dynamics of the physical system following an instantaneous excitation.


© Biraj

## 2.2 The Step Response $s(t)$ and $s[n]$

**Definition: Step Response**

The **Step Response** $s(t)$ (in CT) or $s[n]$ (in DT) is the system output when excited by a unit step function $u(t)$ or $u[n]$ under zero initial state:

$$s(t) \triangleq \mathcal{T}\{u(t)\}, \qquad s[n] \triangleq \mathcal{T}\{u[n]\} \tag{5}$$

## 2.3 Relationship Between Impulse Response and Step Response

Since the unit step is the integral (running sum) of the unit impulse:

$$u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau, \qquad u[n] = \sum_{k=-\infty}^{n} \delta[k] \tag{6}$$

By linearity and time-invariance, the step response is the running integral (or running sum) of the impulse response:

**Fundamental Inter-relationships: Impulse $\leftrightarrow$ Step Response**

**Continuous-Time (CT):**

$$s(t) = \int_{-\infty}^{t} h(\tau)d\tau \iff h(t) = \frac{ds(t)}{dt} \tag{7}$$

**Discrete-Time (DT):**

$$s[n] = \sum_{k=-\infty}^{n} h[k] \iff h[n] = s[n] - s[n-1] = \nabla s[n] \tag{8}$$

where $\nabla$ denotes the backward difference operator.

**Derivation Example: Step Response of First-Order RC Circuit**

Consider a simple low-pass RC series circuit where input $x(t) = v_{in}(t)$ and output $y(t) = v_C(t)$. The differential equation is $RC\frac{dy(t)}{dt} + y(t) = x(t)$.

1. The impulse response is obtained by setting $x(t) = \delta(t)$:

$$h(t) = \frac{1}{RC}e^{-t/RC}u(t)$$

2. The step response $s(t)$ is calculated via running integration for $t \geq 0$:

$$s(t) = \int_{0}^{t} \frac{1}{RC}e^{-\tau/RC}d\tau = \left[-e^{-\tau/RC}\right]_0^t = \left(1 - e^{-t/RC}\right)u(t)$$

3. Differentiating $s(t)$ recovers $h(t)$:

$$\frac{ds(t)}{dt} = \frac{1}{RC}e^{-t/RC}u(t) + (1-e^{-t/RC})\delta(t) = \frac{1}{RC}e^{-t/RC}u(t) + 0 = h(t) \quad (\text{since } 1-e^0 = 0).$$


© Biraj

# 3. The Convolution Operation

## 3.1 Derivation of the Convolution Integral (CT)

Let a continuous-time LSI system $\mathcal{T}$ have impulse response $h(t) = \mathcal{T}\{\delta(t)\}$.

1. Express the input $x(t)$ using the sifting property:

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \tag{9}$$

2. Apply the transformation $\mathcal{T}$ to both sides:

$$y(t) = \mathcal{T}\{x(t)\} = \mathcal{T}\left\{\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau\right\} \tag{10}$$

3. By linearity (additivity over continuum and homogeneity for scalar $x(\tau)$):

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\mathcal{T}\{\delta(t-\tau)\}d\tau \tag{11}$$

4. By shift-invariance, $\mathcal{T}\{\delta(t-\tau)\} = h(t-\tau)$:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \triangleq x(t) * h(t) \tag{12}$$

## 3.2 Derivation of the Convolution Sum (DT)

Similarly, for a discrete-time LSI system where $h[n] = \mathcal{T}\{\delta[n]\}$:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \tag{13}$$

$$y[n] = \mathcal{T}\{x[n]\} = \sum_{k=-\infty}^{\infty} x[k]\mathcal{T}\{\delta[n-k]\} = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \triangleq x[n] * h[n] \tag{14}$$

**Core Formulation: Convolution Integral and Sum**

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau \tag{15}$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] \tag{16}$$

## 3.3 Graphical Interpretation of Convolution

Evaluating convolution graphically consists of four rigorous sequential steps:

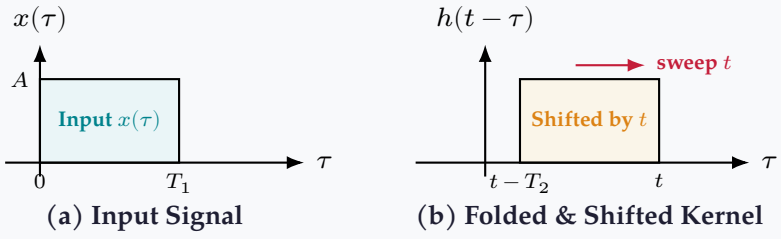
1. **Change of Variable:** Plot $x(\tau)$ and $h(\tau)$ as functions of the dummy variable $\tau$.
2. **Folding (Time Reversal):** Flip $h(\tau)$ about the vertical axis to obtain $h(-\tau)$.


© Biraj

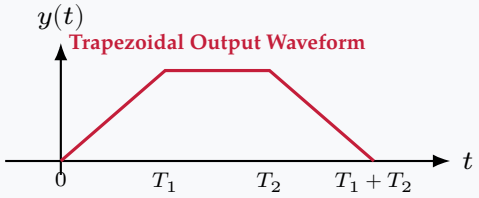
3. **Shifting (Time Delay):** Shift $h(-\tau)$ by time $t$ to obtain $h(t-\tau)$ (right-shift for $t>0$, left-shift for $t<0$).

4. **Multiplication and Integration/Summation:** Multiply $x(\tau)$ and $h(t-\tau)$ pointwise, and evaluate the area under the product curve over the overlapping support interval.

**TikZ Illustration: Graphical Convolution Mechanism**




(a) Input Signal (b) Folded & Shifted Kernel




(c) Resulting Convolution $y(t) = x(t) * h(t)$

## 3.4 Properties of Convolution

| Property | Continuous-Time Formulation | Discrete-Time Formulation |
|---|---|---|
| **Commutative** | $x(t) * h(t) = h(t) * x(t)$ | $x[n] * h[n] = h[n] * x[n]$ |
| **Associative** | $[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$ | $[x[n] * h_1[n]] * h_2[n] = x[n] * [h_1[n] * h_2[n]]$ |
| **Distributive** | $x(t) * [h_1(t) + h_2(t)] = x * h_1 + x * h_2$ | $x[n] * [h_1[n] + h_2[n]] = x * h_1 + x * h_2$ |
| **Shift Property** | $x(t-t_1) * h(t-t_2) = y(t-t_1-t_2)$ | $x[n-n_1] * h[n-n_2] = y[n-n_1-n_2]$ |
| **Identity Element** | $x(t) * \delta(t) = x(t)$ | $x[n] * \delta[n] = x[n]$ |
| **Derivative Property** | $\frac{d}{dt}[x(t) * h(t)] = \frac{dx(t)}{dt} * h(t) = x(t) * \frac{dh(t)}{dt}$ | $\nabla[x[n] * h[n]] = \nabla x[n] * h[n] = x[n] * \nabla h[n]$ |
| **Duration Property** | If lengths are $L_1$ and $L_2$, duration of $y(t)$ is $L_1 + L_2$. | Number of non-zero samples is $N_1 + N_2 - 1$. |

## 3.5 Interconnection of LSI Systems

- **Cascade (Series) Connection:** Equivalent impulse response is the convolution of individual impulse responses:

$$h_{eq}(t) = h_1(t) * h_2(t) = h_2(t) * h_1(t) \tag{17}$$

By the commutative property, the order of cascaded LSI subsystems can be interchanged without altering the overall response.

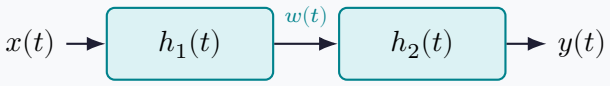

© Biraj

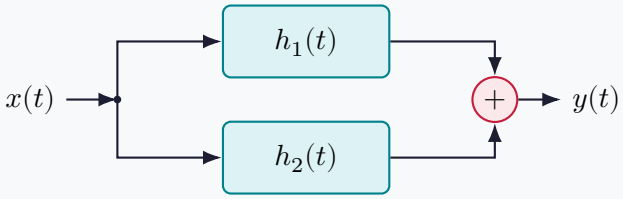
- **Parallel Connection:** Equivalent impulse response is the arithmetic sum of individual responses:

$$h_{eq}(t) = h_1(t) + h_2(t) \tag{18}$$

**TikZ Block Diagrams: Cascade and Parallel LSI Interconnections**

**Cascade Configuration:** $h_{eq}(t) = h_1(t) * h_2(t)$




**Parallel Configuration:** $h_{eq}(t) = h_1(t) + h_2(t)$

# 4. Causality and Stability of LSI Systems

## 4.1 Characterization of Causality

A system is causal if the output at any instant $t_0$ depends only on the input values for $t \leq t_0$.

**Causality Criterion for LSI Systems**

A continuous-time (or discrete-time) LSI system is **causal** if and only if its impulse response is zero for all negative time:

$$\text{CT:} \quad h(t) = 0 \quad \forall t < 0 \tag{19}$$

$$\text{DT:} \quad h[n] = 0 \quad \forall n < 0 \tag{20}$$

**Proof of Causality Condition:** The convolution integral gives:

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau = \int_{-\infty}^{0} h(\tau)x(t-\tau)d\tau + \int_{0}^{\infty} h(\tau)x(t-\tau)d\tau \tag{21}$$

For $\tau < 0$, the term $t-\tau > t$, representing future input values. Thus, for $y(t)$ to be completely independent of future inputs $x(t-\tau)$ $(\tau < 0)$, we must have:

$$h(\tau) = 0 \quad \text{for all } \tau < 0 \tag{22}$$

Under causality, the convolution limits simplify to:

$$y(t) = \int_{0}^{\infty} h(\tau)x(t-\tau)d\tau = \int_{-\infty}^{t} x(\tau)h(t-\tau)d\tau \tag{23}$$

If the input $x(t)$ is also causal $(x(t) = 0$ for $t < 0)$:

$$y(t) = \int_{0}^{t} x(\tau)h(t-\tau)d\tau \quad (t \geq 0) \tag{24}$$


© Biraj

### 4.2 Characterization of BIBO Stability

A system is Bounded-Input Bounded-Output (BIBO) stable if every bounded input $|x(t)| \leq M_x < \infty$ produces a bounded output $|y(t)| \leq M_y < \infty$.

**BIBO Stability Criterion for LSI Systems**

An LSI system is **BIBO stable** if and only if its impulse response is **absolutely integrable** (in CT) or **absolutely summable** (in DT):

$$\text{CT:} \quad \int_{-\infty}^{\infty} |h(t)|dt < \infty \tag{25}$$

$$\text{DT:} \quad \sum_{n=-\infty}^{\infty} |h[n]| < \infty \tag{26}$$

**Rigorous Proof of BIBO Stability Criterion (Sufficiency and Necessity)**

**1. Sufficiency Proof (If $\int |h(t)|dt < \infty \implies$ BIBO Stable):** Let $|x(t)| \leq M_x < \infty$. Taking the magnitude of the output:

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau \right| \leq \int_{-\infty}^{\infty} |h(\tau)| \cdot |x(t-\tau)|d\tau$$
$$\leq M_x \int_{-\infty}^{\infty} |h(\tau)|d\tau$$

Since $\int_{-\infty}^{\infty} |h(\tau)|d\tau = K < \infty$, we have $|y(t)| \leq M_x K < \infty$ for all $t$. Hence, the output is guaranteed bounded.

**2. Necessity Proof (If BIBO Stable $\implies \int |h(t)|dt < \infty$):** Suppose $\int_{-\infty}^{\infty} |h(t)|dt = \infty$. Choose the bounded input:

$$x(t-\tau) = \text{sgn}[h(\tau)] = \begin{cases} +1, & h(\tau) > 0 \\ -1, & h(\tau) < 0 \end{cases}$$

Clearly $|x(t)| = 1 \leq 1$ is bounded. Then at time $t = 0$:

$$y(0) = \int_{-\infty}^{\infty} h(\tau)x(-\tau)d\tau = \int_{-\infty}^{\infty} h(\tau)\text{sgn}[h(\tau)]d\tau = \int_{-\infty}^{\infty} |h(\tau)|d\tau = \infty$$

The output grows unbounded at $t = 0$, violating BIBO stability. Therefore, absolute integrability is both necessary and sufficient.

## 5. System Representation Through Differential Equations


© Biraj

### 5.1 Linear Constant-Coefficient Differential Equations (LCCDE)

A continuous-time lumped parameter LTI system is generally modeled by an $N$-th order LCCDE relating input $x(t)$ to output $y(t)$:

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^{M} b_m \frac{d^m x(t)}{dt^m} \tag{27}$$

where $a_k, b_m$ are real, time-invariant coefficients, and $N \geq M$ for physical realizability.

### 5.2 Complete Solution: Natural and Forced Responses

The total output consists of two components:

$$y(t) = y_h(t) + y_p(t) = y_{zi}(t) + y_{zs}(t) \tag{28}$$

- **Homogeneous Solution $y_h(t)$ (Natural Response):** Solution when input $x(t) \equiv 0$:

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = 0 \implies \sum_{k=0}^{N} a_k \lambda^k = 0 \quad \text{(Characteristic Equation)} \tag{29}$$

  If roots $\lambda_i$ are distinct: $y_h(t) = \sum_{i=1}^{N} C_i e^{\lambda_i t}$.

- **Particular Solution $y_p(t)$ (Forced Response):** The specific response satisfying the non-zero input excitation $x(t)$.
- **Zero-Input Response (ZIR) $y_{zi}(t)$:** Response produced purely by internal stored energy (initial conditions) with zero external input.
- **Zero-State Response (ZSR) $y_{zs}(t)$:** Response produced purely by the input with all initial conditions set to zero: $y_{zs}(t) = x(t) * h(t)$.

### 5.3 Condition of Initial Rest

**Concept: Initial Rest**

An LCCDE system is said to be at **Initial Rest** if the system starts with zero energy storage prior to input application:

$$\text{If } x(t) = 0 \text{ for } t \leq t_0 \implies y(t) = 0 \text{ for } t \leq t_0 \tag{30}$$

Initial rest is the necessary and sufficient condition that guarantees the differential equation system is **Linear**, **Time-Invariant**, and **Causal**.

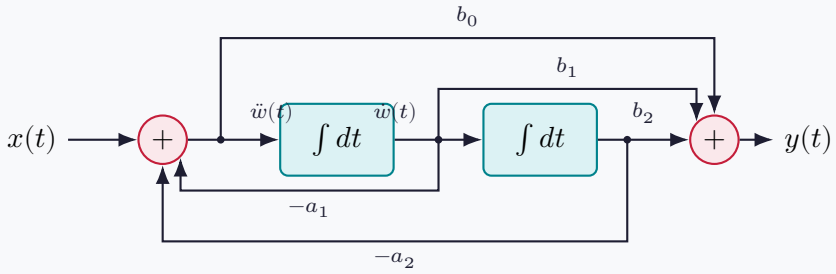
### 5.4 Block Diagram Realization of LCCDE Systems

LCCDEs can be physically synthesized using three elementary operational building blocks:

1. **Integrators ($\int$):** preferred over differentiators because differentiators amplify high-frequency noise.
2. **Adders / Summers ($\sum$):** compute algebraic additions.
3. **Scalers / Multipliers ($a_k, b_m$):** amplify signals by constant gains.


© Biraj

**TikZ Schematic: Direct Form II (Canonic) Realization of 2nd-Order LCCDE**




## 6. Behavior with Aperiodic Convergent Inputs

When an aperiodic convergent input signal (such as an exponentially decaying signal $x(t) = e^{-at}u(t)$, $a > 0$) is applied to an LSI system:

- **Energy Preservation:** If both $x(t)$ and $h(t)$ are $L_1$ and $L_2$ bounded (energy signals), the output $y(t) = x(t) * h(t)$ is also guaranteed to be an energy signal:

$$E_y = \int_{-\infty}^{\infty} |y(t)|^2 dt < \infty \tag{31}$$

- **Convergence to Zero:** Since $\lim_{t\to\infty} x(t) = 0$, for any stable system with $\int |h(t)|dt < \infty$, the output asymptotically decays to zero:

$$\lim_{t\to\infty} y(t) = \lim_{t\to\infty} \int_0^t x(\tau)h(t-\tau)d\tau = 0 \tag{32}$$

- **Transient Decay Rate:** The overall decay envelope is governed by the slowest pole (dominant time constant) among the system and the input excitation.


© Biraj

# Quick Revision & Examination Compendium

Module 2: Linear Shift-Invariant Systems & Convolution

## Key Formulas at a Glance

| Concept / Law | Mathematical Equation |
|---|---|
| **Continuous-Time Convolution** | $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$ |
| **Discrete-Time Convolution** | $y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$ |
| **Step Response from Impulse** | $s(t) = \int_{-\infty}^{t} h(\tau)d\tau, \quad s[n] = \sum_{k=-\infty}^{n} h[k]$ |
| **Impulse Response from Step** | $h(t) = \frac{ds(t)}{dt}, \quad h[n] = s[n] - s[n-1]$ |
| **LTI Causality Condition** | $h(t) = 0 \quad \forall t < 0, \qquad h[n] = 0 \quad \forall n < 0$ |
| **LTI BIBO Stability Condition** | $\int_{-\infty}^{\infty} \lvert h(t)\rvert dt < \infty, \qquad \sum_{n=-\infty}^{\infty} \lvert h[n]\rvert < \infty$ |
| **Cascade Connection** | $h_{eq}(t) = h_1(t) * h_2(t) = h_2(t) * h_1(t)$ |
| **Parallel Connection** | $h_{eq}(t) = h_1(t) + h_2(t)$ |
| **Convolution with Shifted Impulse** | $x(t) * \delta(t - t_0) = x(t - t_0)$ |
| **LCCDE Zero-State Output** | $y_{zs}(t) = x(t) * h(t)$ under initial rest |


© Biraj

## One-Line Definitions for Rapid Review

| Term | Definitive Technical Meaning |
|---|---|
| **Impulse Response** $h(t)$ | The output response of an LTI system when excited by a unit impulse under zero initial conditions. |
| **Step Response** $s(t)$ | The response of an LTI system to a unit step input under initial rest conditions. |
| **Convolution** | A mathematical operator combining two signals to express how the shape of one is modified by the other. |
| **Initial Rest** | The condition where a system has zero internal energy stored prior to the instant an input is applied. |
| **Causal LSI System** | An LTI system whose impulse response is strictly zero for all negative time ($h(t) = 0, t < 0$). |
| **BIBO Stable LSI System** | An LTI system whose impulse response is absolutely integrable ($\int \lvert h(t) \rvert dt < \infty$). |
| **Zero-Input Response (ZIR)** | The system output generated solely by non-zero initial states in the absence of any input. |
| **Zero-State Response (ZSR)** | The system output generated solely by the external input when all initial states are zero. |
| **Canonic Realization** | A block diagram realization containing the minimum possible number of delay or integrator elements ($N$). |

## Frequently Asked / Viva Questions

**Q1. Why does the impulse response $h(t)$ completely characterize an LSI system?**

Because any input $x(t)$ can be represented as an integral of weighted, shifted impulses. By linearity (superposition) and time-invariance, the output is the corresponding integral of weighted, shifted impulse responses. Thus, knowing $h(t)$ determines the response to *any* arbitrary input.

**Q2. How can you compute the impulse response if the step response is known?**

For continuous-time systems, differentiate the step response: $h(t) = \frac{ds(t)}{dt}$. For discrete-time systems, take the first backward difference: $h[n] = s[n] - s[n-1]$.

**Q3. Prove that convolution is commutative: $x(t) * h(t) = h(t) * x(t)$.**

By definition, $x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$. Substituting $v = t - \tau \implies d\tau = -dv$. When $\tau = -\infty, v = +\infty$; when $\tau = +\infty, v = -\infty$. Thus: $\int_{+\infty}^{-\infty} x(t - v)h(v)(-dv) = \int_{-\infty}^{\infty} h(v)x(t-v)dv = h(t) * x(t)$.

**Q4. What is the total duration of the convolution of two signals of lengths $T_1$ and $T_2$?**

In continuous-time, the duration is $T_1 + T_2$. In discrete-time, if signal 1 has $N_1$


© Biraj

**Frequently Asked / Viva Questions (contd.)**

samples and signal 2 has $N_2$ samples, the convolved output has exactly $N_1+N_2-1$ samples.

**Q5. State the necessary and sufficient condition for an LSI system to be causal.**

The impulse response must be zero for all negative time: $h(t) = 0$ for $t < 0$ (or $h[n] = 0$ for $n < 0$).

**Q6. Is the system with impulse response $h(t) = e^{2t}u(t)$ stable? Why?**

No. Evaluating absolute integrability: $\int_{-\infty}^{\infty} |h(t)|dt = \int_0^{\infty} e^{2t}dt = \left[\frac{e^{2t}}{2}\right]_0^{\infty} = \infty$. Since it is not absolutely integrable, it is BIBO unstable.

**Q7. What is the difference between Zero-Input Response (ZIR) and Zero-State Response (ZSR)?**

ZIR is driven purely by initial energy/conditions stored in capacitors/inductors with zero external input. ZSR is driven purely by the applied input with zero initial stored energy. Total response is $y(t) = y_{zir}(t) + y_{zsr}(t)$.

**Q8. Why does the condition of initial rest ensure that a differential equation represents an LTI system?**

Non-zero initial conditions introduce an additive constant independent of the input, violating linearity (homogeneity $y(0) \neq 0$) and causing time-variant behavior. Initial rest enforces zero state prior to excitation, ensuring linearity, time-invariance, and causality.

**Q9. Why are integrators preferred over differentiators in analog computer simulation of LCCDE?**

Differentiators have transfer function $s$, which amplifies high-frequency noise and introduces signal instability. Integrators have transfer function $1/s$, which attenuates high frequencies, acting as low-pass filters that provide stable, noise-free computation.

**Q10. What is the convolution of a signal $x(t)$ with a unit step $u(t)$?**

$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau)u(t-\tau)d\tau = \int_{-\infty}^{t} x(\tau)d\tau$. Convolution with a unit step is equivalent to running integration of the signal from $-\infty$ to $t$.


© Biraj

**Summary Matrix: LSI Characterization Across Domains**

| Property | Continuous-Time (CT) | Discrete-Time (DT) |
|---|---|---|
| **Impulse Input** | **Dirac Delta** $\delta(t)$ | Kronecker Delta $\delta[n]$ |
| **Convolution** | $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$ | $y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$ |
| **Causality** | $h(t) = 0 \quad \forall t < 0$ | $h[n] = 0 \quad \forall n < 0$ |
| **BIBO Stability** | $\int_{-\infty}^{\infty} \lvert h(t)\rvert dt < \infty$ | $\sum_{n=-\infty}^{\infty} \lvert h[n]\rvert < \infty$ |
| **Step Response** $s$ | $s(t) = \int_{-\infty}^{t} h(\tau)d\tau$ | $s[n] = \sum_{k=-\infty}^{n} h[k]$ |
| **Recovery of** $h$ | $h(t) = \frac{ds(t)}{dt}$ | $h[n] = s[n] - s[n-1]$ |
| **Interconnection** | **Cascade:** $h_1 * h_2$**; Parallel:** $h_1 + h_2$ | Cascade: $h_1 * h_2$; Parallel: $h_1 + h_2$ |


© Biraj